SSL Certificate Creation and Configuration for Windows

Table of Contents

Introduction

This guide provides step-by-step instructions for creating SSL certificates using mkcert on Windows and applying them to: 1. A web server (Apache or Nginx) 2. A Windows desktop for browser trust

mkcert creates locally trusted development certificates, allowing you to use HTTPS locally without browser security warnings.

Prerequisites

- Windows 10 or 11 (Pro recommended for system-wide certificate installation)
- PowerShell 5.1 or later
- Administrative privileges (required for system-wide certificate installation and server configuration)
- Web server software (Apache or Nginx) if configuring a web server

Install mkcert on Windows

Option 1: Using Chocolatey (Recommended)

1. Install Chocolatey if not already installed:

   ```
   Set-ExecutionPolicy Bypass -Scope Process -Force; [System.Net.ServicePointManager]::Sec
   ```

2. Install mkcert:

   ```
   choco install mkcert -y
   ```

Option 2: Manual Installation

1. Download the latest Windows binary from mkcert releases
2. Extract the ZIP file to a permanent location (e.g., `C:\tools\mkcert`)
3. Add the location to your PATH environment variable:

   ```
   [Environment]::SetEnvironmentVariable("Path", "$env:Path;C:\tools\mkcert", "User")
   ```

4. Restart PowerShell or your terminal

Verify Installation

```
mkcert --version
```

Generate SSL Certificates

1. Create a directory for certificates:

   ```
   New-Item -ItemType Directory -Path "C:\certs" -Force
   ```

2. Navigate to the certs directory:

   ```
   cd "C:\certs"
   ```

3. Generate certificates for localhost (run as Administrator):

   ```
   mkcert -cert-file localhost.pem -key-file localhost-key.pem localhost 127.0.0.1 ::1
   ```

4. Verify the files were created:

   ```
   Get-ChildItem localhost.pem localhost-key.pem
   ```

5. Install the local CA (required for browser trust):

   ```
   mkcert -install
   ```

Part 1: Install Certificate on a Web Server

Apache Server Configuration

Step 1: Configure Apache for SSL

1. Open the Apache configuration file:

   ```
   notepad "C:\Apache24\conf\httpd.conf"
   ```

2. Ensure the following modules are uncommented:

   ```
   LoadModule ssl_module modules/mod_ssl.so
   LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
   ```

3. Add or modify the following at the end of the file:

   ```
   Listen 443
   ```

4. Save and close the file

Step 2: Create SSL Configuration

1. Create a new SSL configuration file:

   ```
   notepad "C:\Apache24\conf\extra\httpd-ssl.conf"
   ```

2. Add the following configuration (adjust paths as needed):

   ```
   <IfModule ssl_module>
   <VirtualHost *:443>
       ServerName localhost
       ServerAdmin admin@example.com
       DocumentRoot "C:/Apache24/htdocs"
       ErrorLog "logs/error.log"
       CustomLog "logs/access.log" common

       SSLEngine on
       SSLCertificateFile "C:/certs/localhost.pem"
       SSLCertificateKeyFile "C:/certs/localhost-key.pem"

       <FilesMatch "\.(cgi|shtml|phtml|php)$">
           SSLOptions +StdEnvVars
       </FilesMatch>
       <Directory "C:/Apache24/cgi-bin">
           SSLOptions +StdEnvVars
       </Directory>

       BrowserMatch "MSIE [2-5]" \
           nokeepalive ssl-unclean-shutdown \
           downgrade-1.0 force-response-1.0
       # MSIE 7 and newer should be able to use keepalive
       BrowserMatch "MSIE [17-9]" ssl-unclean-shutdown
   </VirtualHost>
   </IfModule>
   ```

3. Save and close the file

Step 3: Include SSL Configuration

1. Open the main configuration file again:

   ```
   notepad "C:\Apache24\conf\httpd.conf"
   ```

2. Add this line at the end of the file:

   ```
   Include conf/extra/httpd-ssl.conf
   ```

3. Save and close the file

Step 4: Start Apache with SSL

1. Open Command Prompt as Administrator
2. Navigate to Apache directory:

   ```
   cd C:\Apache24\bin
   ```

3. Start Apache:

   ```
   httpd.exe
   ```

4. Verify Apache is running:

   ```
   netstat -ano | findstr 443
   ```

5. Test in browser: https://localhost

Nginx Server Configuration

Step 1: Configure Nginx for SSL

1. Open the Nginx configuration file:

   ```
   notepad "C:\nginx\conf\nginx.conf"
   ```

2. Add the following server block inside the `http` block:

   ```
   server {
       listen 443 ssl;
       server_name localhost;

       ssl_certificate "C:/certs/localhost.pem";
       ssl_certificate_key "C:/certs/localhost-key.pem";

       location / {
           root C:/nginx/www;
           index index.html index.htm;
       }

       error_page 500 502 503 504 /50x.html;
       location = /50x.html {
           root C:/nginx/html;
       }
   }
   ```

3. Save and close the file

Step 2: Start Nginx with SSL

1. Open Command Prompt as Administrator
2. Navigate to Nginx directory:

```
cd C:\nginx
```

3. Start Nginx:

```
start nginx.exe
```

4. Verify Nginx is running:

```
netstat -ano | findstr 443
```

5. Test in browser: https://localhost

Part 2: Install Certificate in Windows Desktop for Browsers

Install in Windows Trust Store

Method 1: Using certutil (PowerShell - Administrator)

```
# Find the mkcert root CA location
$caroot = mkcert -CAROOT

# Install the CA in the Trusted Root Certification Authorities store
certutil -addstore -f "ROOT" "$caroot\rootCA.pem"
```

Method 2: Interactive Installation (GUI)

1. Open the mkcert directory in Windows Explorer:

```
explorer.exe "$env:USERPROFILE\.local\share\mkcert"
```

2. Double-click `rootCA.pem`
3. Click "Install Certificate…"
4. Select "Local Machine" (requires admin) → Next
5. Choose "Place all certificates in the following store"
6. Click "Browse" → Select "Trusted Root Certification Authorities" → OK
7. Click Next → Finish

Verify Installation

```
# Check if the certificate is in the Trusted Root store
certutil -store ROOT | Select-String -Pattern "mkcert"
```

Chromium-Based Browser Configuration

Chromium-based browsers (Chrome, Edge, Brave, etc.) use the Windows Certificate Store by default. However, you may need to:

Clear HSTS Cache

1. Open a new tab and navigate to:

   ```
   chrome://net-internals/#hsts
   ```

2. In the "Delete domain security policies" section:
   - Enter `localhost`
   - Click "Delete"
3. Restart the browser

Alternative: Use Incognito Mode Chromium-based browsers in Incognito mode don't use the HSTS cache, so certificates will work immediately.

Firefox Configuration

Firefox has its own certificate store and requires separate configuration:

Method 1: Automatic Installation (Recommended) When you run `mkcert -install`, it should automatically install the CA in Firefox. Verify with:

```
mkcert -CAROOT
```

Method 2: Manual Installation

1. Open Firefox
2. Go to: `about:preferences#privacy`
3. Scroll down to "Certificates" → "View Certificates"
4. Go to the "Authorities" tab
5. Click "Import…"
6. Navigate to: `%USERPROFILE%\.local\share\mkcert\rootCA.pem`
7. Check "Trust this CA to identify websites" → OK
8. Restart Firefox

Troubleshooting

Certificate Not Trusted

1. **Verify certificate installation**:

   ```
   certutil -store ROOT | Select-String -Pattern "mkcert"
   ```

2. **Reinstall the certificate** (if missing):

   ```
   $caroot = mkcert -CAROOT
   certutil -addstore -f "ROOT" "$caroot\rootCA.pem"
   ```

Browser Shows Security Warning

1. **Close all browser windows completely** (check system tray for running instances)
2. **Use Incognito mode** to test
3. **Clear HSTS cache** as shown above
4. **Restart your computer** (sometimes required for certificate changes to take effect)

Server Not Using Certificate

1. **Verify file paths** are correct in your server configuration
2. **Check server logs** for SSL-related errors
3. **Verify certificate permissions**:

   ```
   icacls "C:\certs\localhost.pem"
   icacls "C:\certs\localhost-key.pem"
   ```

4. **Restart the server** after configuration changes

Complete Certificate Reset

```
# Remove old certificates
Remove-Item -Recurse -Force "C:\certs"

# Uninstall mkcert CA
mkcert -uninstall

# Remove mkcert directory
Remove-Item -Recurse -Force "$env:USERPROFILE\.local\share\mkcert"

# Reinstall mkcert
mkcert -install

# Generate new certificates
cd "C:\certs"
mkcert -cert-file localhost.pem -key-file localhost-key.pem localhost 127.0.0.1 ::1

# Reinstall in Windows Trust Store
$caroot = mkcert -CAROOT
certutil -addstore -f "ROOT" "$caroot\rootCA.pem"
```

References

- mkcert GitHub Repository
- Apache SSL/TLS Configuration
- Nginx SSL/TLS Configuration

- Windows certutil Documentation